

Provable Differential and Linear Bounds for a Butterfly-Diffusion Permutation: The XRBD Layer and Krakken-2048

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Abstract

We introduce the XOR-Rotation Butterfly Diffusion (XRBD) layer, a bijective diffusion primitive for wide-state permutations, and prove tight active-S-box bounds for Krakken-2048, the 2048-bit permutation that uses it. The base design without XRBD was described in an earlier preprint [2]; the present work adds the XRBD layer and, crucially, replaces the earlier empirical security estimates with proofs. Using an exact byte-lane Mixed-Integer Linear Programming (MILP) model solved to a closed optimality gap, we prove a minimum of **229** active S-boxes over the eight-round permutation, for *both* differential and linear trails, bounding any single characteristic by 2^{-1374} . The bound is established directly at every round from one to eight (values 5, 37, 69, 101, 133, 165, 197, 229), not extrapolated. We isolate the contribution of the XRBD layer by solving the model with and without it: the SPN core alone admits a cheap lane-aligned trail (27 active S-boxes at two rounds), which XRBD provably destroys, raising the count to 37 at two rounds and 69 at three, with the margin growing per round. We also verify, by complete enumeration on the constants as implemented, the foundational properties the bounds rely on: the maximal MDS branch number (9), the S-box differential uniformity and linear bias ($DP = \text{corr}^2 = 2^{-6}$), and the bijectivity of all permutation layers. We reproduce, for completeness, the bijectivity and optimal-depth proofs for XRBD. We are explicit about scope: the bounds concern single characteristics; the linear bound coincides with the differential one by wide-trail symmetry rather than via an independent bit-level model; and a division-property analysis and external cryptanalysis remain open.

1 Introduction

The wide-trail strategy [6, 7] bounds the probability of any single differential or linear characteristic of an SPN by counting the minimum number of S-boxes the characteristic must activate. A diffusion layer added to such a design is cryptographically useful only if it does not lower—and ideally raises—that count. The base permutation Krakken-2048, an SPN-ARX hybrid *without* a butterfly diffusion layer, was described in an earlier preprint [2], whose provable differential resistance was weak: its SPN core admits a cheap lane-aligned characteristic. In this paper we introduce the XOR-Rotation Butterfly Diffusion (XRBD) layer—a bijective $\lceil \log_2 n \rceil$ -stage butterfly network for wide states—and prove that adding it raises the permutation’s active-S-box bound dramatically, while retaining the wide-trail guarantees of the SPN core.

Contributions. We introduce the XRBD layer and prove it is a bijection with optimal mixing depth (Section 2). We then construct an exact byte-lane MILP model of the complete round function and solve it to a *closed* branch-and-bound optimality gap, so each reported count is a proof of a lower bound rather than a heuristic estimate or the output of a time-limited search. Our results are:

1. **A proven eight-round bound.** The minimum number of active S-boxes over the eight-round permutation is 229, for both differential and linear trails, giving a bound of 2^{-1374} on any single characteristic. Every per-round value from one to eight is proven directly (Table 3).
2. **Isolation of the XRBD contribution.** Solving the model with and without XRBD shows that XRBD strictly increases the active-S-box count, by a margin that grows with depth (+10 at two rounds, +14 at three). The SPN core alone admits a trail riding the MDS layer at its exact branch number along a single byte-lane; XRBD’s cross-word rotations destroy it.
3. **Exhaustively verified foundations.** The MDS branch number (9), the S-box differential and linear properties (2^{-6} each), and the bijectivity of the permutation layers are verified by complete enumeration on the implemented constants.

Reproducible provers, foundational proof scripts, a verified inverse permutation, and known-answer test vectors accompany this work. The empty-string digest (256-bit output, rate 160 bytes) is:

e1c646acd24c1211dc58f38fe7cb95ec
9ca41b7c57cff233e6faf1fac8b2e0dd

Relation to prior work. The base permutation without XRBD is described in [2], which we cite for the underlying SPN-ARX components and recall only as needed (Section 3). The XRBD layer, its structural proofs (Section 2), and all of the proven active-S-box bounds (Sections 4–6) are contributions of the present work.

What we do not claim. The bounds concern single characteristics; differential clusters and linear hulls are not quantified (the 2^{-1374} margin makes them implausible but this is not proven). The linear bound coincides with the differential one by wide-trail symmetry rather than by an independent bit-level mask model. The ARX layer is modelled with addition as activity-spreading (carries ignored), keeping the counts sound lower bounds. Security of any mode built on the permutation, and side-channel resistance, are out of scope, as is independent external cryptanalysis, which we regard as the essential next step.

2 The XRBD layer

XRBD operates on an n -word state ($n = 32$ for Kracken-2048) as a butterfly (FFT-topology) network of bijective crossover steps. Its graph structure is classical; its cryptographic content is the crossover step below, whose bijectivity and optimal mixing depth we now establish.

Theorem 1 (Bijectivity of the XRBD crossover step). *Let the crossover at stage k between words x, y be $x' = x \oplus y$, $y' = y \oplus \text{RotL}(x', r_k)$. This map is a bijection for any word length w and any rotation $r_k \in \mathbb{Z}_w$.*

Proof. It suffices to exhibit a unique inverse. From the second equation $y = y' \oplus \text{RotL}(x', r_k)$; since x' is an output, $\text{RotL}(x', r_k)$ is determined, so y is recovered uniquely. From the first equation $x = x' \oplus y = x' \oplus y' \oplus \text{RotL}(x', r_k)$, which is determined by the outputs. Hence the map is invertible, therefore bijective. $\square$

Theorem 2 (Optimal dependency expansion). *After $\log_2 n$ stages of XRBD_n , every output word depends on every input word; and any diffusion network built from pairwise-mixing operations requires at least $\log_2 n$ stages to achieve such complete dependency. XRBD attains this lower bound.*

Proof. Completeness. Let $x_i^{(k)}$ be word i after stage k , with $x_i^{(-1)} = \mathbf{S}[i]$, and at stage k (distance $\delta_k = 2^k$), $x_i^{(k)} = x_i^{(k-1)} \oplus x_{i \oplus \delta_k}^{(k-1)}$ and $x_{i \oplus \delta_k}^{(k)} = x_{i \oplus \delta_k}^{(k-1)} \oplus \text{RotL}(x_i^{(k)}, r_k)$, for indices whose k -th bit is 0. We show by induction on k that $x_j^{(k)}$ depends on all inputs whose index agrees with j on bit positions $> k$. *Base* ($k = 0$): stage 0 mixes indices differing only in bit 0, so $x_j^{(0)}$ depends on inputs agreeing with j on bits $1, \dots, \log_2 n - 1$. *Step:* at stage k , $x_j^{(k)}$ combines $x_j^{(k-1)}$ and $x_{j \oplus \delta_k}^{(k-1)}$; by hypothesis these depend on inputs agreeing with j resp. $j \oplus \delta_k$ on bits $\geq k$, and since $j, j \oplus \delta_k$ differ only at bit k , the union covers all inputs agreeing with j on bits $> k$. (For indices with bit k equal to 1, the update uses the second equation of the crossover step; both cases produce the same dependency union, since the second output depends on the first output $x_i^{(k)}$ which already incorporates both inputs $x_i^{(k-1)}$ and $x_{i \oplus \delta_k}^{(k-1)}$.) After the final stage $k = \log_2 n - 1$ the restriction is empty, so every output word depends on all n inputs. *Optimality.* Let $D(s)$ be the maximum number of inputs influencing any word after s stages; $D(0) = 1$. A pairwise operation unions two dependency sets, so $D(s) \leq 2D(s-1)$, giving $D(s) \leq 2^s$. Complete dependency needs $D(s) \geq n$, hence $2^s \geq n$, i.e. $s \geq \log_2 n$. XRBD reaches complete dependency in exactly $\log_2 n$ stages (5 for $n = 32$), attaining the bound. $\square$

3 The Krakken-2048 round function (recap)

We recall only what the bound analysis needs; the full specification, including all constants of the base components, is in [2]. The state is 2048 bits as 32 lanes of 64 bits, viewed as 8 columns $\times$ 4 rows. One round applies, in order: Theta (column-parity mixing); MDS, a circulant $\text{GF}(2^8)$ matrix with branch number 9; Rho (lane rotations); Pi (lane permutation); Chi, the nonlinear layer applying the ABYSSAL 8-bit S-box (differential uniformity 4, maximum squared correlation 2^{-6}) with a cross-lane coupling; XRBD (Section 2); Pressure, a 64-bit ARX intra-column step; Iota (SHAKE128-derived round constants); and InkCloud (a lane shuffle). The permutation iterates $R = 8$ rounds.

4 Proven active-S-box bounds

Model. We encode trail propagation as a byte-lane MILP in the style introduced by Mouha et al. [3]: a binary variable marks each potentially active S-box, the MDS layer is constrained so that the number of active input and output lanes in each row is either 0 or at least the branch number 9, and the rotation, permutation, butterfly, and ARX layers propagate activity exactly (the ARX additions are modelled as activity-spreading, ignoring carries, which keeps the resulting counts sound *lower* bounds). Minimising the total number of active S-boxes subject to a nonzero input difference yields a provable lower bound on any single characteristic. We solve with the open-source solver SCIP (version 9.1.0, [1]) and report only instances whose branch-and-bound optimality gap is closed to *exactly* 0.00%, as reported by the solver. Table 1 reports the solve statistics for all eight instances; every instance closed to a gap of exactly 0.00%.

Table 1 reports the solve statistics for all eight instances.

Our model operates at the byte-lane level; bit-level MILP models [4, 5] can capture additional structure but are not required for the lane-counting bounds established here.

Isolating XRBD. We solve two models: the SPN core (Theta $\rightarrow$ MDS $\rightarrow$ Rho $\rightarrow$ Pi $\rightarrow$ Chi) and the complete round. The SPN core admits a two-round trail with only 27 active S-boxes, which rides the MDS layer at its exact branch number along a single byte-lane. Inserting XRBD destroys this trail: the two-round minimum rises to 37 and the three-round minimum from 55 to 69 (Table 2). The margin grows with depth (+10 then +14), confirming that the butterfly’s cross-word rotations couple the byte-lanes the SPN core leaves separable.

Rounds	Min. S-boxes	Time (s)	Nodes	Gap
1	5	140.92	264	0.00%
2	37	244.50	617	0.00%
3	69	286.35	444	0.00%
4	101	670.08	1101	0.00%
5	133	1,596.72	1090	0.00%
6	165	1,899.47	1267	0.00%
7	197	5,031.85	3129	0.00%
8	229	2,570.74	1746	0.00%

Table 1: SCIP branch-and-bound solve statistics for all eight instances. Primal and dual bounds coincide in every case, confirming exact optimality. All instances were solved on a Dell Latitude 5400 (Intel Skylake, 16 GB RAM). The non-monotone solve time (round 7 harder than round 8) is a normal MILP phenomenon: problem difficulty depends on the search tree structure, not solely on problem size.

Rounds	SPN core	SPN core + XRBD	Gain
1	5	5	+0
2	27	37	+10
3	55	69	+14

Table 2: Effect of the XRBD layer on the minimum active-S-box count (proven, optimality gap closed). At one round the counts coincide, since the S-boxes are counted at the Chi input, before XRBD acts; from two rounds on, XRBD strictly increases the count, by a margin that grows with depth.

Bounds for the complete permutation. Carried to the full permutation, the minimum active-S-box count is established directly at every round (Table 3).

Main result. The eight-round minimum is 229 active S-boxes, proven directly (not extrapolated). Hence no eight-round differential characteristic has probability above 2^{-1374} . From two rounds onward each added round contributes exactly 32 active S-boxes; we record this regularity but do not depend on it, since the eight-round value is itself proven.

5 Linear cryptanalysis

At the byte-lane activity level the linear-trail model is the transpose of the differential model. The MDS branch number is invariant under transposition, and the remaining layers propagate activity identically in both directions, so the minimum number of linearly-active S-boxes equals the differential count at every round. With the S-box maximum squared correlation 2^{-6} (Section 6), the eight-round bound on any single linear characteristic is likewise 2^{-1374} . We present this as a wide-trail coincidence: the linear bound shares the structure of the differential model rather than being derived from an independent bit-level mask analysis.

6 Exhaustively verified foundations

The bounds rest on three properties, each verified by complete enumeration on the implemented constants.

Rounds	Min. active S-boxes	DP / corr^2 bound
1	5	2^{-30}
2	37	2^{-222}
3	69	2^{-414}
4	101	2^{-606}
5	133	2^{-798}
6	165	2^{-990}
7	197	2^{-1182}
8	229	2^{-1374}

Table 3: Proven minimum active-S-box counts for the complete round function (MILP, optimality gap closed), established directly at every round. The differential and linear counts coincide at every round (Section 5); the DP/corr² column is $(2^{-6})^{\text{count}}$.

- **MDS branch number 9.** All 12,869 square submatrices of the circulant matrix are nonsingular over $\text{GF}(2^8)$ (reduction polynomial **0x11D**), equivalent to branch number 9.
- **S-box.** The ABYSSAL S-box is a bijection with differential uniformity 4 ($\text{DP} = 2^{-6}$) and maximum linear bias 16 (maximum squared correlation 2^{-6}), confirmed by the full 256×256 DDT and LAT.
- **Permutation layers.** The Pi and InkCloud word permutations are bijections (the InkCloud multiplier 7 is invertible modulo 32, inverse 23). A full inverse permutation has been implemented and verified to round-trip with the forward permutation at all round counts.

7 Empirical distinguisher battery

Independently of the proven bounds, the permutation was tested against standard distinguishers over 10^6 independent trials per test unless otherwise noted; all results below are *empirical* (they exclude a detectable distinguisher at the sampled scale) and corroborate but do not extend the MILP bounds above.

Avalanche. A single input-bit flip changes a mean of 0.500 ± 0.001 of the output bits from round 1 onward (ideal: 0.5; measured over 10^6 random states and bit positions). Full 2048-bit diffusion is achieved by round 1.

Rotational symmetry. Comparing $P(\text{rotate}(X, k))$ with $\text{rotate}(P(X), k)$ gives mean Hamming distance ≈ 1024 bits at all tested rotation offsets $k \in \{1, \dots, 2047\}$ and all round counts (10^4 trials per offset). No rotational relation was detected.

Integral / cube. A saturated input byte (all 256 values) produces no balanced output coordinate at round 1 or beyond (XOR sum nonzero in all 256 output bytes, verified exhaustively). A cube-sum test with dimension 12 (2^{12} inputs, 500 random cube positions) shows the zero-sum property vanishing by round 2; no cube sum was zero at round 3 or beyond ($p < 0.01$ false-positive threshold).

Zero-sum. Neither a forward zero-sum nor a two-sided zero-sum distinguisher (the latter using the verified inverse permutation) was found over 10^5 trials against the full permutation.

Algebraic degree. The exact algebraic degree over sampled variable sets of size ≤ 20 reaches near-maximal value by round 2, consistent with the degree saturation analysis of [2].

8 Open problems

The following remain open and would strengthen or independently confirm the analysis: an independent bit-level linear model (rather than the transpose coincidence); a carry-aware ARX model

and a division-property (MILP) degree and integral analysis; differential-cluster and linear-hull estimates; security analysis of sponge and authenticated-encryption modes built on the permutation; and, most importantly, independent external cryptanalysis. We provide the differential and linear MILP provers, the foundational proof scripts, the verified inverse permutation, and known-answer test vectors to support such analysis.

9 Conclusion

We have proven, via exact gap-closed MILP, that the eight-round Krakken-2048 permutation has at least 229 active S-boxes against both differential and linear single-characteristic cryptanalysis, bounding any such characteristic by 2^{-1374} , and we have isolated the XRBD layer as the source of the improvement over the SPN core. The bounds are proven rather than estimated, the underlying constants are verified by enumeration, and the scope of what is and is not established is stated explicitly. The design now warrants independent cryptanalysis, for which complete reproducible artifacts are provided.

References

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A Complete Specification

This appendix gives a self-contained specification of the eight-round Krakken-2048 permutation sufficient to implement and verify all results in this paper. All integer arithmetic is unsigned modulo 2^{64} ; all indices are zero-based.

A.1 State representation

The state is an array $\mathbf{S}[0 \dots 31]$ of 64-bit words (2048 bits total), logically arranged as an 8×4 matrix: column c occupies words $\mathbf{S}[4c], \mathbf{S}[4c + 1], \mathbf{S}[4c + 2], \mathbf{S}[4c + 3]$ for $c \in \{0, \dots, 7\}$.

A.2 Abyssal S-box

The ABYSSAL S-box $\mathcal{S} : \{0, \dots, 255\} \rightarrow \{0, \dots, 255\}$ is defined by $\mathcal{S}(x) = A \cdot \text{inv}(x) \oplus c$ where $\text{inv}(x)$ is multiplicative inversion in $\text{GF}(2^8)/\langle x^8 + x^4 + x^3 + x^2 + 1 \rangle$ (polynomial `0x11D`), A is the 8×8 circulant matrix with seed byte `0x13`, and $c = \text{0xA5}$. The lookup table is given in Table 4.

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0x	A5	B6	DE	F7	18	37	8C	C1	89	DA	1E	85	31	F0	97	77
1x	41	14	E8	C8	8A	04	B5	69	1D	2B	0F	2C	4E	19	CC	79
2x	D7	4D	7D	43	03	3A	13	92	32	D9	75	DF	AD	81	C3	F1
3x	F9	A7	E2	35	02	DD	61	A2	50	E1	09	C5	E3	71	CB	99
4x	9C	B1	23	86	3B	93	24	E9	F6	B4	6A	66	FE	7A	3E	28
5x	6E	F2	9B	F8	3F	2A	98	10	A1	FB	45	36	64	57	8F	72
6x	8B	29	56	FD	F4	A4	ED	A6	76	EB	6B	4A	C7	5E	26	D0
7x	5F	CA	87	52	01	16	67	B9	74	4B	CF	D2	60	2F	49	6F
8x	39	1C	5D	53	E6	3C	C6	7F	EA	E5	BE	00	65	88	83	E4
9x	0C	38	2D	80	B0	AB	44	84	08	0D	B8	51	9A	2E	91	68
Ax	40	0A	FC	82	BA	CE	0B	FA	1A	5B	62	22	C9	3D	8D	06
Bx	55	D5	78	AE	27	9D	9E	AF	B7	4F	DC	9F	42	A3	BC	15
Cx	B2	DB	11	A9	5C	E7	7B	EF	FF	C2	25	EE	73	F5	D6	48
Dx	4C	21	70	D1	30	54	A0	B3	94	07	58	AA	96	1B	1F	0E
Ex	D8	17	E0	BB	46	6C	AC	A8	05	7E	8E	33	C4	D4	59	BD
Fx	BF	F3	20	34	90	CD	EC	63	47	95	12	6D	D3	5A	C0	7C

Table 4: Abyssal S-box (hexadecimal; row = high nibble, column = low nibble). Properties: nonlinearity 112, differential uniformity 4, algebraic degree 7, zero fixed points.

A.3 Round function steps

One round applies the following eight steps in order.

Theta. For each column c , compute parity $P_c = \mathbf{S}[4c] \oplus \mathbf{S}[4c + 1] \oplus \mathbf{S}[4c + 2] \oplus \mathbf{S}[4c + 3]$, then for each row $r \in \{0, 1, 2, 3\}$:

$$\mathbf{S}[4c + r] \oplus= \text{rotr}(P_{(c+7) \bmod 8}, 1) \oplus P_{(c+1) \bmod 8}.$$

Tentacle (MDS). For each row r , let $v_c = \mathbf{S}[4c + r]$. Each word is treated as 8 bytes; multiplication is in $\text{GF}(2^8)/\langle \text{0x11D} \rangle$. The output is

$$w_c = \bigoplus_{i=0}^7 \text{GF_mul}(v_{(c+i) \bmod 8}, K_i), \quad K = [01, 01, 04, 01, 08, 05, 02, 09]_{\text{hex}}.$$

This circulant matrix is the Whirlpool MDS matrix [7] and has branch number 9.

Rho. Each word $\mathbf{S}[i]$ is rotated left by ρ_i bits:

$$\rho = [32, 1, 62, 28, 36, 44, 15, 61, 6, 19, 24, 55, 3, 10, 43, 17, 25, 39, 41, 59, 47, 8, 56, 14, 18, 35, 21, 33, 2, 49, 22, 51].$$

Pi. Words are permuted: word at position (x, r) (i.e. index $4x + r$) moves to column $(x + 3r) \bmod 8$, row r unchanged.

Chi. For each row r and each pair $p \in \{0, 1, 2, 3\}$, let $a = \mathbf{S}[8p + r]$, $b = \mathbf{S}[8p + 4 + r]$:

$$a' = \mathcal{S}(a \oplus \text{rotl}(b, 32)), \quad b' = \mathcal{S}(b \oplus \text{rotl}(a', 32)).$$

XRBD. Five stages $k = 0, \dots, 4$ with distances $\delta_k = 2^k$ and rotation constants $[r_0, r_1, r_2, r_3, r_4] = [13, 23, 37, 41, 53]$. At stage k , for each index i with bit k equal to 0:

$$x' = \mathbf{S}[i] \oplus \mathbf{S}[i \oplus \delta_k], \quad y' = \mathbf{S}[i \oplus \delta_k] \oplus \text{rotl}(x', r_k), \quad \mathbf{S}[i] \leftarrow x', \quad \mathbf{S}[i \oplus \delta_k] \leftarrow y'.$$

Pressure. For each column c , let $(a, b, d_c, d) = (\mathbf{S}[4c], \mathbf{S}[4c + 1], \mathbf{S}[4c + 2], \mathbf{S}[4c + 3])$ (additions modulo 2^{64}):

$$\begin{aligned} a &+= d_c \oplus (d_c \gg 17), & b &+= d \oplus (d \gg 17), \\ d_c &+= a \oplus (a \ll 31), & d &+= b \oplus (b \ll 31). \end{aligned}$$

Store $\mathbf{S}[4c] \leftarrow a$, $\mathbf{S}[4c + 1] \leftarrow \text{rotl}(b, 7)$, $\mathbf{S}[4c + 2] \leftarrow d_c$, $\mathbf{S}[4c + 3] \leftarrow \text{rotl}(d, 19)$.

Iota. XOR round constants: $\mathbf{S}[i] \oplus= \text{RC}[r][i]$ for $i \in \{0, \dots, 31\}$, where $\text{RC}[r]$ is the r -th row of the 8×32 constant table. Constants are derived by squeezing SHAKE-128 on the UTF-8 string "Krakken-2048 Abyssal v1 - Primary " and decoding the output as 256 little-endian 64-bit integers; any zero value is replaced by `0xDEADBEEFCAFEBABE`.

InkCloud. Each word $\mathbf{S}[i]$ is first rotated left by 11 bits, then scattered: $\text{dest}[(7i) \bmod 32] \leftarrow \text{src}[i]$ for all $i \in \{0, \dots, 31\}$. The multiplier 7 is invertible modulo 32 (inverse 23), so this is a bijection on the word indices.

A.4 Sponge parameters and padding

State width	2048 bits (256 bytes)
Rate	160 bytes (1280 bits)
Capacity	96 bytes (768 bits)
Rounds	8
Padding	Keccak multi-rate: <code>0x06</code> at position $ m \bmod 160$, <code>0x80</code> at byte 159 (merged as <code>0x86</code> when the message fills the final block exactly)

A.5 Known-answer test vectors

All vectors use a 256-bit (32-byte) output digest computed by the scalar reference implementation (`krakken_hash_scalar`), rate 160 bytes, 8 rounds.

The empty-string vector matches the value stated in Section 1 and serves as the primary interoperability check. All five vectors were verified to produce identical output from both the scalar reference implementation (`krakken_hash_scalar`) and the AVX2 implementation (`krakken_hash_avx2`), confirming bit-exact equivalence between the two code paths across all boundary conditions (short input, rate−1, exact rate multiples).

Remark 1. *The empty-string digest in the earlier preprint [2] (`0c513a0b...`) differs from the value above because that version used 10 rounds. All vectors in this paper correspond to the current 8-round design with the XRBD layer.*

Input	Digest (hex)
" " (empty, 0 bytes)	e1c646acd24c1211dc58f38fe7cb95ec 9ca41b7c57cff233e6faf1fac8b2e0dd
"abc" (3 bytes)	5062b998511e180f476907438a27183f fcd657031d141b3659f9f32611d9fe7c
0x00×159 (rate−1 bytes)	6a7449835760839e3d2d629fc839eae6 376348c8339fd0f34b8c75264083bc93
0x00×160 (one full rate block)	2d87727b66f7d07827cb1ca1a705da69 4361b935bfb41b3a6b78cec93a837e13
0x00×320 (two full rate blocks)	290f56f6f29cb0faf5969404fab2290e db9aa01fd2d4acfc00b29cf1c4da5ec

Table 5: Known-answer test vectors for Krakken-2048 (8 rounds, scalar reference implementation, 256-bit output). The 159-byte vector exercises the merged padding byte 0x86; the 160- and 320-byte vectors exercise exact and double rate-block boundaries.